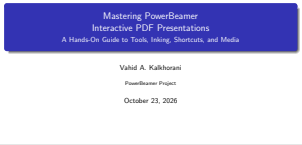
# Mastering PowerBeamer
# Interactive PDF Presentations

## A Hands-On Guide to Tools, Inking, Shortcuts, and Media

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PowerBeamer Project

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**PRESENTER NOTES (Slide 1)**:

- Welcome to the interactive PowerBeamer tutorial deck!
- Notice how this note displays in the presenter window without appearing on the audience screen.
- Press **L** to activate the laser pointer and practice guiding attention.
- Use the left/right arrow keys or spacebar to advance slides.

# Step 1: Annotation & Inking Suite

## Live Pen & Highlighter Tools

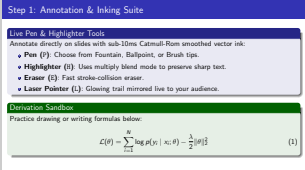
Annotate directly on slides with sub-10ms Catmull-Rom smoothed vector ink:

- **Pen (P)**: Choose from Fountain, Ballpoint, or Brush tips.
- **Highlighter (H)**: Uses multiply blend mode to preserve sharp text.
- **Eraser (E)**: Fast stroke-collision eraser.
- **Laser Pointer (L)**: Glowing trail mirrored live to your audience.

## Derivation Sandbox

Practice drawing or writing formulas below:

$$\mathcal{L}(\theta) = \sum_{i=1}^{N} \log p(y_i \mid x_i; \theta) - \frac{\lambda}{2}\|\theta\|_2^2 \tag{1}$$




**SPEAKER NOTES (Slide 2)**:

- Switch to the **Pen tool (P)** and circle the regularization term $\frac{\lambda}{2}\|\theta\|_2^2$.
- Try changing ink colors using the color palette in the top bar.
- Press **Cmd+Z** / **Ctrl+Z** to test multi-level undo.
- Press **N** or **+** to spawn a blank whiteboard whenever you need extra writing space!

# Step 2: Embedded Audio Support

## Interactive Audio Playback

Click the trigger box below to play the embedded 440 Hz reference tone:

**[AUDIO] Click to Play 440 Hz Reference Audio**
(Embedded via `\movie` from `multimedia.sty`)

- Native audio decoder with zero-latency playback.
- Controls appear on the presenter HUD and seamlessly sync with the presentation.
- Click anywhere on the box or press play to test your speaker output.



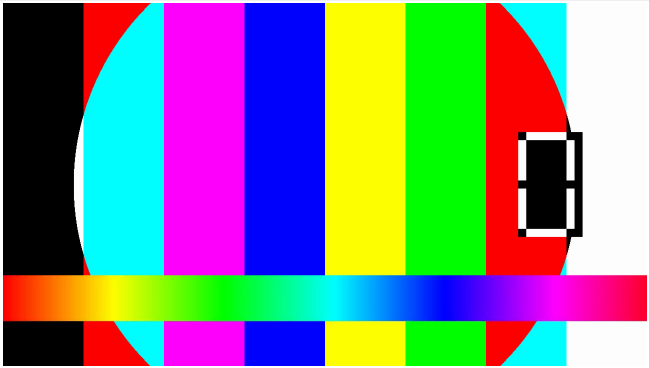

**SPEAKER NOTES (Slide 3)**:

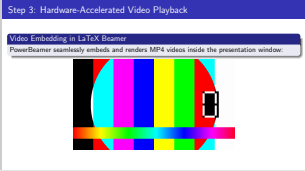
- Click the audio box above to verify media playback through your speakers.
- PowerBeamer extracts media clips defined with LaTeX `\movie` automatically.
- Check the audio volume and mute controls in the presenter controls toolbar.

# Step 3: Hardware-Accelerated Video Playback

## Video Embedding in LaTeX Beamer

PowerBeamer seamlessly embeds and renders MP4 videos inside the presentation window:





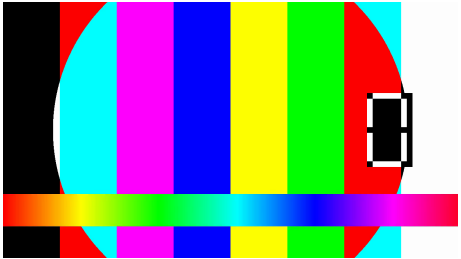

**SPEAKER NOTES (Slide 4)**:

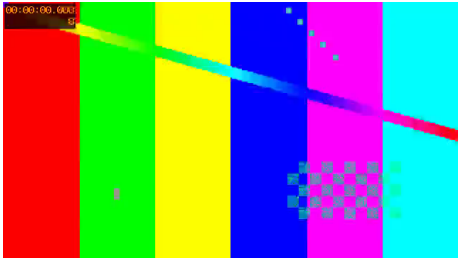
- Click on the video container to start playback.
- Note the hardware-accelerated decode with zero frame dropping.
- You can seek along the timeline scrub bar at the bottom.
- Audience and presenter screens stay perfectly synchronized.

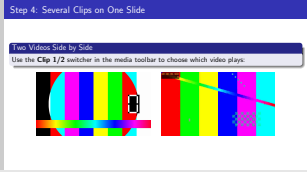
# Step 4: Several Clips on One Slide

## Two Videos Side by Side

Use the **Clip 1/2** switcher in the media toolbar to choose which video plays:








**SPEAKER NOTES (Slide 5)**:

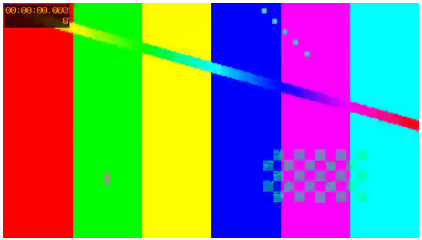
- The selected clip is outlined in red on the presenter screen.
- Switching clips stops the current one and resets the timeline.

# Step 5: Audio Narration Next to a Video

## Mixed Media

The audio clip comes first in the source, so it is selected by default:

**[AUDIO] 440 Hz Narration Track**






**SPEAKER NOTES (Slide 6)**:

- Press play for the narration, then switch to Clip 2/2 for the video.

# Step 6: Essential Shortcuts & Dual Display

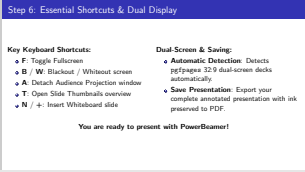
**Key Keyboard Shortcuts:**

- **F**: Toggle Fullscreen
- **B** / **W**: Blackout / Whiteout screen
- **A**: Detach Audience Projection window
- **T**: Open Slide Thumbnails overview
- **N** / **+**: Insert Whiteboard slide

**Dual-Screen & Saving:**

- **Automatic Detection**: Detects `pgfpages` 32:9 dual-screen decks automatically.
- **Save Presentation**: Export your complete annotated presentation with ink preserved to PDF.

**You are ready to present with PowerBeamer!**




**SPEAKER NOTES (Slide 7)**:

- Wrap up the tutorial walkthrough.
- Press **T** to view the slide thumbnail grid and rearrange or navigate between slides.
- Test saving the annotated PDF from the top action menu.